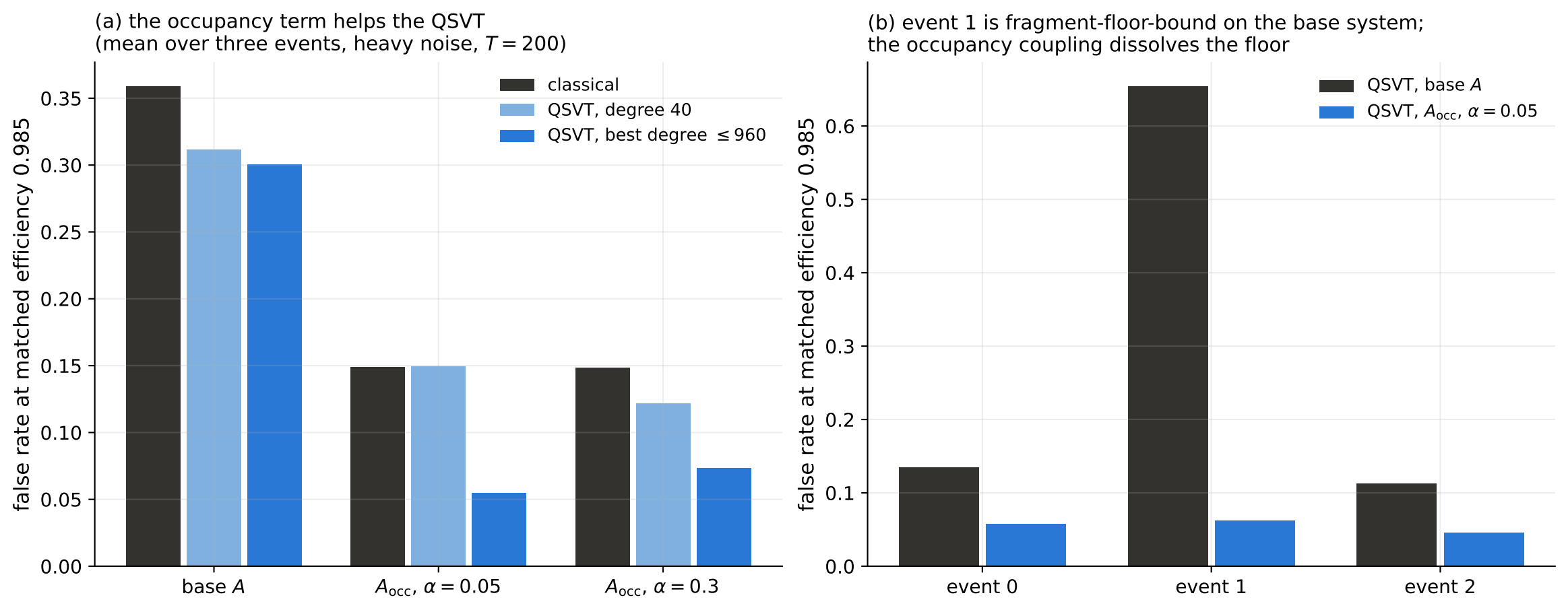
(a) the occupancy term helps the QSVT
(mean over three events, heavy noise, $T = 200$)
false rate at matched efficiency 0.985
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
classical
QSVT, degree 40
QSVT, best degree ≤ 960
base A
A_{occ}, $\alpha = 0.05$
A_{occ}, $\alpha = 0.3$
(b) event 1 is fragment-floor-bound on the base system;
the occupancy coupling dissolves the floor
false rate at matched efficiency 0.985
0.0
0.1
0.2
0.3
0.4
0.5
0.6
QSVT, base A
QSVT, A_{occ}, $\alpha = 0.05$
event 0
event 1
event 2